Telecom Paris **4AI10 - Machine Learning**
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TP : Principal Component Analysis

- Discovering Python -

To start with or find a reminder for the Python language you may use the following materials:

- http://www.python.org
- http://scipy.org
- http://www.numpy.org
- http://scikit-learn.org/stable/index.html
- http://www.loria.fr/~rougier/teaching/matplotlib/matplotlib.html
- http://jrjohansson.github.io/

- Questions -

1) A random vector X in $\mathbb{R}^d$ is said to be generated from a **multivariate Student-*t* distribution** with ν degrees of freedom with center $\mu \in \mathbb{R}^d$ and scatter matrix $\Sigma \in \mathbb{R}^{d\times d}$ if it can be represented as:

$$X = \boldsymbol{\mu} + \sqrt{\frac{\nu}{W_\nu}} Z\,,$$

where W_ν is a variable following chi-squared distribution with ν degrees of freedom and Z follows a d-variate normal distribution centered in the origin and with the covariance matrix Σ $\mathcal{N}(0_d, \Sigma)$, W_ν and Z being independent.

For more information on chi-squared distribution an a related Gamma distribution see: https://en.wikipedia.org/wiki/Chi-squared_distribution and https://en.wikipedia.org/wiki/Gamma_distribution.

Program a function **rmvt** that generates a data set from a multivariate Student-t distribution and takes as arguments center μ, scatter matrix Σ, number of degrees of freedom ν, and number of points to generate.

Draw a data set from the bivariate Student-t distribution with the following sets of parameters

- $\mathcal{N}(\left[\begin{smallmatrix}0\\0\end{smallmatrix}\right], \left[\begin{smallmatrix}1&1\\1&4\end{smallmatrix}\right])$ with 5 degrees of freedom, which we will call MVS1 in the sequel;
- $\mathcal{N}(\left[\begin{smallmatrix}2\\2\end{smallmatrix}\right], \left[\begin{smallmatrix}1&1\\1&4\end{smallmatrix}\right])$ with 3 degrees of freedom, which we will call MVS2 in the sequel;
- $\mathcal{N}(\left[\begin{smallmatrix}2\\2\end{smallmatrix}\right], \left[\begin{smallmatrix}4&4\\4&16\end{smallmatrix}\right])$ with 1 degree of freedom, which we will call MVS3 in the sequel.

each containing 100 points and plot them.

(Student-t distribution with 1 degree of freedom is called the Cauchy distribution.)

2) Generate low-rank noisy data.

For this, first generate 1000 observations from a Student-t5 distribution centered in the origin with the covariance matrix $\left[\begin{smallmatrix}1&1&1\\1&4&4\\1&4&9\end{smallmatrix}\right]$. Further, place this data into the Euclidean space of dimension 5 using the following projection matrix:

$\left[\begin{smallmatrix}1&1&1&1&1\\1&-1&1&-1&0\\0&-1&-1&0&2\end{smallmatrix}\right]^\top$

normalized by columns, not to change the distances between the points (check this). Further, shift the data set by $[1\,2\,3\,4\,5]^\top$.

Finally, add random normal noise to each point:
$\mathcal{N}\left(\left[\begin{smallmatrix}0\\0\\0\\0\\0\end{smallmatrix}\right], \left[\begin{smallmatrix}0.025 & 0 & 0 & 0 & 0\\ 0 & 0.025 & 0 & 0 & 0\\ 0 & 0 & 0.025 & 0 & 0\\ 0 & 0 & 0 & 0.025 & 0\\ 0 & 0 & 0 & 0 & 0.025\end{smallmatrix}\right]\right)$.

The final data set shall be of low-rank 3 (with noise) saved in the matrix 1000×5.

3) Principal Component Analysis (PCA)

   Run the principal component analysis on this data set. Print explained variances, singular values, and full principal components.

   Plot three plots of pairs of first three principal components:
   $\mathrm{PC}_1 \times \mathrm{PC}_2$, $\mathrm{PC}_1 \times \mathrm{PC}_3$, and $\mathrm{PC}_2 \times \mathrm{PC}_3$.

   Further, plot the three following bivariate configurations of principal components:
   $\mathrm{PC}_4 \times \mathrm{PC}_5$, $\mathrm{PC}_1 \times \mathrm{PC}_4$, and $\mathrm{PC}_2 \times \mathrm{PC}_5$.

   Comment on your observation.

4) PCA via Singular Value Decomposition (SVD)

   Compute SVD of the covaraince matrix.

   Compare with the principal components. Do they coincide?

5) Contaminated data

   Replace 10% of last observations in the data set by 100 generated from (standard) Cauchy distribution.

   Plot first two coordinates. What is your conclusion?

6) PCA on contaminated data.

   Compute PCA on these contaminated data and print explained variances, singular values, and full principal components. Do principal components coincide with those from Tasks 3 and 4?

   Now compute the Minimum Covariance Determinant estimate on the same data set.

   Does the mean estimate look reasonable?

   Run SVD on the (corrected) MCD-estimated covariance matrix. What can you say about principal components (do they coincide with thos from Tasks 3 and 4)?